



The Relationship Between Short Video-Based Social Media Use, Dopamine Signaling, and Depressive Symptoms

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Abstract

This narrative review examines the relationship between short-form video-based social media platforms (TikTok, Instagram Reels, YouTube Shorts), dopamine signaling, and depressive symptoms. The synthesis of current literature from 2020–2026 indicates that these platforms create a digital ecosystem that continuously activates the brain's reward system through algorithmic design features such as infinite scrolling, autoplay, and variable reward mechanisms. The findings suggest that unpredictable content streams trigger the reward prediction error mechanism, activating dopaminergic reward circuits and potentially leading to compulsive usage behaviors.

From a mental health perspective, there is a generally positive but non-linear relationship between short-form video use and depressive symptoms. This relationship is shaped not only by usage duration but also by usage patterns, social comparison, sleep disturbances, and individual psychological factors.

Although some studies report potential beneficial effects, the evidence is largely cross-sectional and self-reported, limiting causal interpretations, and leaving neurobiological mechanisms largely at a theoretical level.

In conclusion, this review highlights a multifactorial and context-dependent relationship between short-form video use, dopamine signaling, and depressive symptoms, emphasizing the need for future longitudinal and experimental studies.

Keywords

Short-form video platforms; social media; dopamine signaling; reward system; reward prediction error; short-form video addiction; depressive symptoms; mental health

1. Introduction

With the acceleration of digitalization, social media platforms have significantly changed individuals' habits of accessing information, communicating, and entertainment. One of the most remarkable reflections of this transformation is the rapid global proliferation of short video-based social media platforms such as TikTok, Instagram Reels, and YouTube Shorts. Supported by advanced algorithms, these platforms aim to keep users' attention on the platform for extended periods through personalized content recommendations, the infinite scroll feature, and a continuously refreshed stream of short videos. Thus, unlike traditional

content consumption, a new digital consumption model has emerged in which short-duration but highly stimulating content predominates.

It is considered that the intensive stream of stimuli provided by short video-based social media platforms may affect neurobiological processes associated with the brain's reward system. In particular, it has been suggested that the dopaminergic system, which plays an important role in reward anticipation, motivation, and learning processes, may play an active role in the use of these platforms because of variable reward mechanisms and the continuous presentation of new content. This situation is associated with behavioral reinforcement processes that may contribute to users remaining on the platform for longer periods; in recent years, the possible effects of these mechanisms on mental health have been investigated increasingly. The widespread global use of these platforms has made the investigation of their potential neurobiological and psychological effects even more important. Current data regarding the prevalence of the use of major short video platforms are presented in Figure 1.

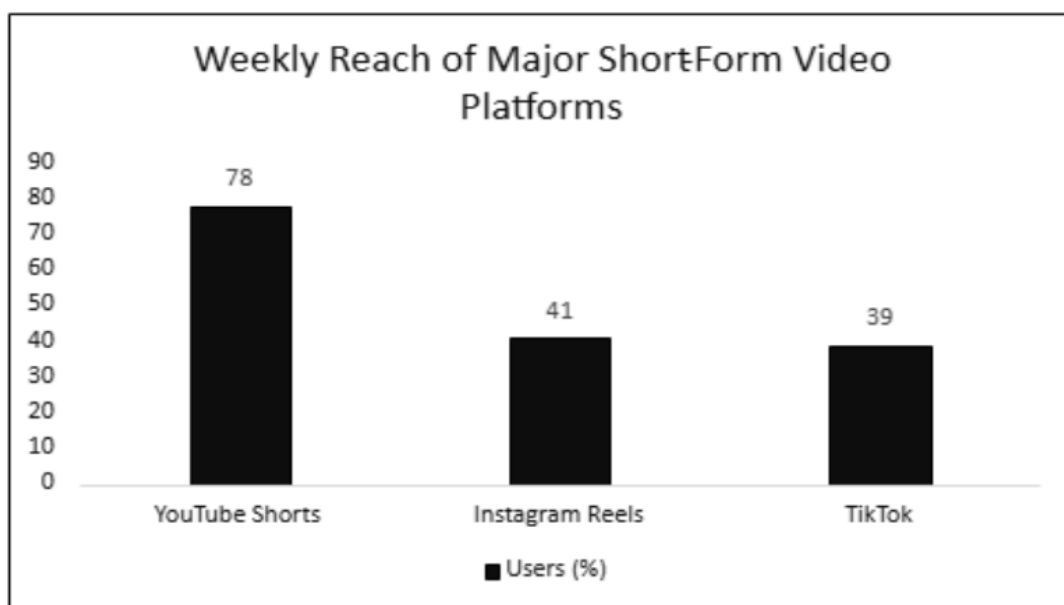


Figure 1. Weekly usage rates of major short video platforms. The data were compiled from the Ampere Analysis (2025) report.

In the literature, the number of studies reporting that problematic social media use may be associated with various mental health indicators, such as depressive symptoms, anxiety, loneliness, low life satisfaction, and sleep disturbances, is increasing. However, the biological and behavioral mechanisms underlying the relationship between the effects of short video-based social media use on dopamine signaling and depressive symptoms have not yet been fully elucidated. Furthermore, since the majority of existing studies are cross-sectional and correlational in nature, significant uncertainties remain regarding the direction and causality of this relationship.

The aim of this narrative review is to evaluate the relationship between short video-based social media use, dopamine signaling, and depressive symptoms in light of the current literature; to synthesize the current findings from a holistic neurobiological and psychological

perspective; and to identify knowledge gaps in this field while providing a conceptual framework for future research.

2. Methods

This study is a narrative review prepared to evaluate the relationship between short video-based social media use, dopamine signaling, and depressive symptoms in light of the current scientific literature. The narrative review method enables the integration of findings obtained from different research designs, the evaluation of the existing body of knowledge on the subject from a holistic perspective, and the identification of prominent trends and knowledge gaps in the literature. Accordingly, studies investigating the potential effects of short video-based social media use on neurobiological mechanisms and mental health were comprehensively reviewed and synthesized thematically. All figures presented in this study were originally prepared for the purpose of synthesizing and visualizing the scientific evidence based on the data obtained through the literature review.

The literature search was conducted using electronic databases commonly used in the fields of biomedicine, psychology, and health sciences. In this context, a planned search strategy was applied in the PubMed/MEDLINE, Scopus, Web of Science, PsycINFO, and Google Scholar databases. In order to reflect the current state of the subject, peer-reviewed scientific studies published primarily between 2020 and 2026 were reviewed. In addition, seminal theoretical studies and classical references explaining dopamine signaling, the reward system, and the neurobiological basis of depression were also included where appropriate.

English keywords were used in the literature search. The main search terms included "short-form video", "short video platforms", "TikTok", "Instagram Reels", "YouTube Shorts", "social media", "dopamine", "dopaminergic signaling", "reward system", "reward pathway", "depression", "depressive symptoms", "anhedonia", "mental health", and "behavioral addiction". To identify relevant studies more comprehensively, these keywords were searched in different combinations using Boolean operators (AND, OR). In addition, the reference lists of the retrieved articles were examined, and other studies related to the subject were identified using the snowballing method.

Specific criteria were taken into consideration during the study selection process. Original studies conducted on human participants, systematic reviews, meta-analyses, and experimental studies considered important to the subject were included in the evaluation. In addition, fundamental neuroscience studies explaining the mechanisms underlying the dopaminergic reward system and depressive symptoms were reviewed to contribute to the development of the theoretical framework. In contrast, studies addressing only general social media use without separately evaluating short video-based platforms, non-peer-reviewed publications, conference abstracts, letters to the editor, studies for which the full text was not

available, and publications not directly related to the research topic were excluded from the evaluation.

The selected studies were examined in detail and classified thematically in accordance with the aim of the study. The findings were evaluated under three main themes: (i) usage characteristics of short video-based social media platforms and user behaviors, (ii) dopamine signaling, the reward system, and potential neurobiological mechanisms, and (iii) depressive symptoms and other mental health outcomes. Findings obtained from different research designs were comparatively evaluated, and common findings, conflicting findings, and methodological differences were interpreted from a critical perspective. Due to the heterogeneity of the included studies, a quantitative meta-analysis was not performed; instead, a narrative synthesis approach was adopted, and the available evidence was evaluated within a holistic framework.

In interpreting the findings included in this review, not only the statistical results but also the methodological characteristics of the studies, sample structures, measurement instruments, and potential limitations were taken into consideration. Thus, this review aimed to comprehensively evaluate the relationship between short video-based social media use, dopamine signaling, and depressive symptoms in light of the current scientific evidence, to identify the prominent trends in the literature, and to provide a conceptual framework for future empirical research.

3. Literature Findings

3.1. Prevalence and Usage Characteristics of Short Video-Based Social Media Use

The reviewed literature shows that short video-based social media platforms (TikTok, Instagram Reels, and YouTube Shorts) are currently used widely and intensively across different age groups. It has been reported that these platforms have become an important part of daily life, particularly among university students and adolescents, that the duration of use has been gradually increasing, and that problematic patterns of use have emerged in some individuals. Studies have revealed that, due to the rapidly consumable nature of short video content, users are exposed to prolonged exposure, and that this situation may affect attention, time perception, and cognitive control processes [1,4].

One of the main factors determining the widespread use of short video platforms is the algorithmic structure of these systems. The reviewed studies show that platforms such as TikTok, Instagram Reels, and similar platforms continuously provide new and engaging content by analyzing user behavior through personalized content recommendation systems. This algorithmic structure optimizes the content feed based on users' previous interactions

and thereby increases the length of time that individuals remain on the platform. In particular, the infinite scroll and autoplay features create an uninterrupted content consumption cycle by reducing users' conscious decision-making processes [5].

The literature also emphasizes that short video use is closely associated with habitual scrolling behavior. In the study conducted by Alruwaili (2025), it was reported that scrolling intensity had negative effects on attention and working memory and increased cognitive fatigue [4]. Similarly, Qin et al. (2022) stated that, on platforms such as TikTok, system quality (algorithmic structure and user interface) is more influential than information quality, and that this situation strengthens the flow experience and increases addiction-like usage behaviors [5].

Another important factor explaining the high prevalence of short video use is that individuals use these platforms not only for entertainment purposes but also for emotional regulation, escape, and leisure-time use. In the study conducted by Chen and Wang (2024), it was reported that short video applications could have an effect in reducing depressive symptoms, particularly among certain user groups (e.g., functionally dependent individuals), but that this effect was shaped through interpersonal relationships and social interaction [3]. This finding reveals that short video use does not produce the same effect in every individual and may lead to different outcomes depending on the motivation for use and individual characteristics.

In addition, studies conducted on young adults and adolescents show that there are significant relationships between short video addiction and demographic variables. In particular, it has been reported that the levels of short video addiction are higher among female individuals, individuals in younger age groups, and individuals with lower academic performance. It has also been stated that attention and memory processes play a mediating role in this addictive behavior and that impairments in cognitive functions may increase the duration of use [1].

Overall, the literature shows that short video-based social media platforms are not only a means of content consumption but also constitute a complex digital ecosystem shaped by algorithmic design, user interaction, and cognitive processes. Design features such as infinite scroll, autoplay, and personalized content feeds increase the amount of time users spend on the platform and provide a basis for the development of habitual patterns of use. These findings reveal that the widespread use of short video platforms is not merely a social trend but also a process supported by cognitive and behavioral mechanisms.

3.2. Dopamine Signaling and the Reward Mechanism

Dopamine is a chemical that functions as both a neurotransmitter and a neuromodulator in the central nervous system and plays a critical role particularly in reward processing, motivation, attention, and the reinforcement of behavior. The primary function of dopamine is not to directly produce "pleasure," as is commonly thought, but rather to regulate motivation toward

reward and reward anticipation. Therefore, the dopamine system is associated not so much with the reward itself as with the probability of obtaining the reward and the ability of this probability to guide behavior [6,11].

From a neurobiological perspective, the main components of this system are located in the ventral tegmental area (VTA) and the substantia nigra. Dopamine released from these regions is transmitted primarily to reward circuits associated with the nucleus accumbens, the limbic system, and the prefrontal cortex. This network regulates the motivational responses of the individual to environmental stimuli, thereby enabling the development of reward-oriented behaviors. Thus, the dopamine system functions as a critical reinforcement mechanism in the survival and learning processes of the organism.

The functioning of the reward system is largely shaped by the difference between the expected reward and the received reward. This difference is defined in the neuroscience literature as the "reward prediction error." Ishino et al. demonstrated that the dopamine system generates an error signal that encodes not only the presence of a reward but also whether the reward was expected or unexpected [7]. Dopamine activity increases when a reward is unexpected, whereas the failure of an expected reward to occur leads to a reduction in the dopaminergic response. This mechanism forms the basis of learning processes and enables the continuous updating of behavior.

Interestingly, the dopamine system is sensitive not only to the presence of rewards but also to active signals generated in response to the absence of rewards. Dopaminergic responses that occur in cases of unexpected reward omission enable the organism to reorganize its behavior. This demonstrates that the brain is not a passive waiting system but rather a dynamic learning mechanism that continuously adapts [7].

When this neurobiological structure is combined with the variable reward mechanism, a powerful behavioral reinforcement system emerges. In a variable reward structure, the timing and content of the reward are unpredictable. This unpredictability keeps the dopamine system in a continuous state of anticipation, thereby increasing the likelihood that the behavior will be repeated. Social media platforms structurally utilize this mechanism.

On short video-based social media platforms, users encounter the potential for a new reward with every scroll. De et al. stated that this process creates not only a behavioral but also a neurobiological feedback loop [6]. Artificial intelligence-supported algorithms analyze user behavior and present the content with the highest likelihood of engagement, thereby continuously reproducing reward anticipation. This situation leads to a reinforcement mechanism that can be described as a "personalized dopamine loop."

Pedrouzo's studies show that TikTok and similar short video platforms intensively activate the dopaminergic system because of their rapid and variable reward structure [9]. Since users encounter the possibility of a different reward with each new piece of content, the expectation that "the next content may be more interesting" remains continuously active. This contributes to prolonged viewing time and the development of compulsive behavior.

According to Rodrigues, digital platforms may lead to changes in dopaminergic functioning by affecting dopamine transporters (DAT) [11]. These changes are particularly associated with reduced attentional control and compulsive seeking behavior toward rapidly presented stimuli. In the literature, this phenomenon is sometimes conceptualized as the "TikTok brain."

The prefrontal cortex plays a critical role in decision-making and cognitive control processes. Intensive digital media use may weaken the inhibitory functions of the prefrontal cortex and make individuals more sensitive to short-term rewards [6]. In parallel, increased activity in the limbic system may lead to heightened emotional reactivity and weakened behavioral control.

The findings of Dresch-Langley show that digital addiction affects not only behavior but also biological rhythms and neurotransmitter balance. In particular, changes in the dopamine and serotonin systems are reported to play a determining role in sleep patterns, motivation, and mood [8]. These findings indicate that changes in the reward system may affect not only immediate behaviors but also long-term psychophysiological processes.

In conclusion, dopamine signaling is a central mechanism that constitutes the neurobiological basis of short video-based social media use. When reward prediction error, the variable reward structure, and the algorithmic content feed are considered together, a cycle in which the brain's reward system is continuously activated emerges. This cycle constitutes the fundamental neurobiological framework explaining the compulsive use, reduced attention span, and continuous content-seeking behaviors observed on short video platforms [6,7,9,10,11].

3.3. The Relationship Between Short Video Use and Depressive Symptoms

The widespread use of short video-based social media platforms has become an important pattern of digital behavior affecting not only cognitive and neurobiological processes but also mental health indicators. The literature shows that intensive and uncontrolled social media use, in particular, is significantly associated with depressive symptoms, anxiety levels, and overall psychological well-being. This relationship is generally addressed within the framework of problematic social media use, and a bidirectional interaction model has been proposed [12].

Findings from systematic reviews show that there is a consistent, but not always linear, relationship between social media use and depression and anxiety. In particular, rather than the duration of use alone being the determining factor, factors such as the nature of use (active or passive use), the level of emotional engagement, and nighttime use are reported to be more strongly associated with depressive symptoms [12]. This indicates that short video

platforms produce psychological effects not only through "how much they are used" but also through "how they are used."

Studies conducted specifically on TikTok and similar short video platforms reveal that these platforms have a high addiction potential and exhibit a strong association with depressive symptoms, particularly among young users. The systematic review by Jain et al. showed that problematic TikTok use increases symptoms of anxiety and depression and that this effect is more pronounced in individuals under 24 years of age [14]. In addition, female users and individuals with higher levels of neuroticism were reported to be at greater risk [14].

Studies conducted in the adolescent population show that TikTok use disorder has indirect effects on cognitive functions through depression, anxiety, and stress. In this context, depression is reported to be not only an outcome but also a mediating mechanism interacting with cognitive effects such as memory impairment and attentional deficits [13]. This suggests that there is a reciprocal interaction cycle between mental health and cognitive functions.

Another important mechanism related to depressive symptoms is social comparison and emotional exhaustion. The continuous exposure to highly stimulating and idealized content on short video platforms may lead individuals to evaluate their self-perception negatively. This situation provides a basis for the development of depressive symptoms such as low self-esteem, feelings of loneliness, and loss of motivation [12,14].

In addition, the relationship between the dopaminergic reward system and depression is also important in this context. Previous neurobiological studies have shown that excessive stimulation of the reward system may lead to desensitization over time and that this may increase depressive symptoms associated with anhedonia (the inability to experience pleasure) [6,11]. This suggests that, while short video use initially provides reward-based reinforcement, it may contribute to reduced emotional satisfaction in the long term.

Sleep disturbances also emerge as an important mediating factor associated with depressive symptoms. Increased digital media use, particularly during nighttime hours, may disrupt the circadian rhythm and strengthen depressive symptoms by causing changes in the dopamine-serotonin balance [3]. This mechanism constitutes the biological bridge between digital addiction and mood disorders.

Overall, the literature does not attribute the effect of short video platforms on depressive symptoms to a single cause; rather, it explains this effect through the interaction of cognitive, emotional, and neurobiological processes. When social comparison, excessive stimulation of the reward system, attentional impairments, and sleep disturbances are considered together, short video use appears to constitute a multifactorial structure that increases the risk of depression [12,13,14].

In conclusion, although the relationship between short video use and depressive symptoms is strong, it occurs through complex and indirect mechanisms. This relationship is particularly more pronounced among young individuals and high-risk groups, indicating that the effects

of digital media use on mental health should be evaluated not only in terms of duration but also in terms of content, patterns of use, and individual psychological characteristics.

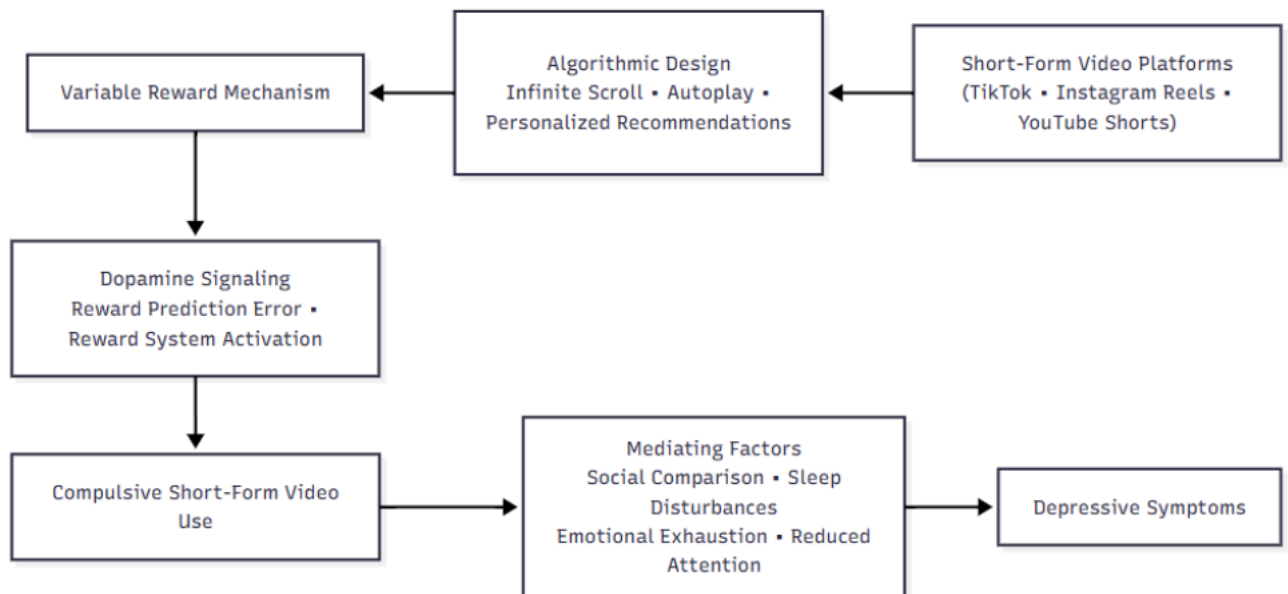


Figure 2. Conceptual model illustrating the relationship between the algorithmic design of short video-based social media platforms, dopamine signaling, and depressive symptoms.

3.4. Consistent and Contradictory Findings in the Current Literature

In accordance with the conceptual model summarized in Figure 2, it is evident that the relationship between the algorithmic design of short video-based social media platforms, dopamine signaling, and depressive symptoms has a multifactorial structure. When the literature on the neurobiological and psychological effects of short video-based social media use is examined, it is observed that although there is an overall tendency toward a strong association, the findings are not entirely consistent and contain notable contradictions in certain aspects. In particular, methodological and theoretical differences among studies are evident with respect to the dopamine system, addiction mechanisms, and mental health outcomes.

One group of studies argues that short video platforms have a direct and strong effect on the dopaminergic reward system. These studies state that social media interactions activate reward centers such as the ventral tegmental area and the nucleus accumbens, and that the variable reward structure and algorithmic content feed continuously stimulate dopamine release, leading to addiction-like behaviors [6,9,11]. In particular, the rapid and unpredictable presentation of content on platforms such as TikTok is considered a mechanism that continuously maintains reward anticipation, thereby increasing the reinforcement of behavior.

In contrast, some studies emphasize that neurobiological inferences are limited because they rely on indirect indicators rather than directly measuring dopamine-related effects. For example, some studies assume changes in the dopamine system but do not provide direct biological measurements (e.g., neuroimaging or neurochemical data). This indicates that the concept of "increased dopamine" remains a theoretical model in some studies [6]. Therefore, while some dopamine-based explanations in the literature are supported by strong experimental evidence, others are based primarily on a theoretical framework.

A similar inconsistency also exists with respect to mental health outcomes. Many studies have demonstrated that short video and social media use shows a significant positive association with depression, anxiety, and stress levels [12,13,14]. In particular, problematic social media use is defined as a risk factor that increases depressive symptoms through factors such as loneliness, social comparison, and sleep disturbances. However, some studies also state that this relationship is not linear and that the duration of use alone is not a determining factor [12].

In particular, some findings indicate that short video applications do not produce negative effects under all circumstances and may, under certain conditions, also have positive effects on psychological well-being by increasing social connectedness. For example, it has been reported that a reduction in depressive symptoms may be observed when social interaction and interpersonal relationships are strengthened [3]. This indicates that the effects of digital media are context-dependent and cannot be explained by a one-directional "harm model."

When evaluated from a methodological perspective, a substantial proportion of the contradictions in the literature arises from differences in study designs. While cross-sectional studies are insufficient for establishing causality, studies based on self-report scales carry the risk of measurement bias. In addition, different age groups, cultural contexts, and patterns of platform use also cause variability in the findings [14].

In addition, some studies make inferences based on behavioral indicators rather than directly examining neurobiological processes. This creates uncertainty in the interpretation of findings related to the dopamine system and reward mechanisms. In particular, although mechanisms such as reward prediction error and variable reward provide strong theoretical explanations, direct biological validation is not available in all studies [7].

Overall, when the literature is evaluated, there is broad consensus that short video use is associated with addiction-like behaviors and mental health indicators; however, there are considerable differences regarding the strength, direction, and mechanisms of this relationship. These differences arise from the fact that neither neurobiological explanations nor psychological models have yet been fully integrated.

In conclusion, although the current literature provides a strong theoretical foundation for explaining the effects of short video-based social media use, it also contains certain contradictions, particularly due to the lack of evidence regarding causality, limitations in measurement accuracy, and the scarcity of longitudinal data. This indicates that future studies

should incorporate more objective measurements, experimental designs, and long-term longitudinal follow-up studies.

4. Discussion

This narrative review examined the relationship between short video-based social media use, dopamine signaling, and depressive symptoms within a multidimensional framework. The findings obtained show that short video platforms not only establish behavioral habits but may also produce neurobiological effects through the reward system, cognitive control mechanisms, and mood regulation processes. However, the literature contains both strong correlational findings and important contradictions arising from methodological differences.

Overall, the studies demonstrate that short video platforms are capable of continuously activating the dopaminergic reward system through their variable reward structure and algorithmic content feed. This phenomenon is explained by the repeated stimulation of reward circuits such as the ventral tegmental area and the nucleus accumbens and constitutes a neurobiological basis for the reinforcement of behavior. However, the fact that this mechanism is explained in most studies through behavioral and theoretical models rather than direct biological measurements renders some dopamine-based interpretations indirect.

In particular, reward prediction error and variable reward mechanisms emerge as central theoretical frameworks for explaining short video use. These models propose that the dopamine system functions more strongly when rewards are unpredictable and that this accelerates behavioral learning. However, the fact that these theoretical explanations are addressed to different extents across different studies creates conceptual heterogeneity in the literature. This indicates that caution should be exercised when interpreting findings related to the dopamine system.

When evaluated in terms of mental health outcomes, short video use is generally found to be positively associated with depressive symptoms and anxiety. In particular, problematic patterns of use may contribute to an increase in depressive symptoms through factors such as social comparison, emotional exhaustion, and sleep disturbances. However, some studies emphasize that this relationship is not linear and that the nature of use (active/passive use and the level of emotional engagement) is at least as important as the duration of use.

At this point, one of the most important contradictions in the literature is that short video use may involve both risks and potential benefits. Some studies show that, in certain contexts, short video platforms may support psychological well-being by increasing social connectedness. This finding demonstrates that the effects of digital media cannot be explained by a one-directional "harm model" and that contextual factors play a critical role.

When evaluated from a methodological perspective, a substantial proportion of the contradictions in the literature arises from study designs. The inability of cross-sectional studies to establish causality, the risk of bias associated with self-report-based scales, and the use of different sample groups reduce the comparability of the findings. In addition, the fact that most neurobiological inferences regarding the dopamine system are based on theoretical modeling rather than direct measurement limits the generalizability of the findings.

In addition, structural differences among platforms (TikTok, Instagram Reels, and YouTube Shorts) also contribute to the variability of findings in the literature. Since each platform differs in its algorithmic structure, content delivery speed, and mode of user interaction, short video use may not be considered a single homogeneous variable. This increases the importance of platform-specific analyses in future studies.

On the other hand, the fact that neurobiological and psychological models have not yet been fully integrated also emerges as an important limitation. Although dopamine signaling, reward prediction error, and prefrontal cortex control mechanisms provide strong explanations, it is still not fully clear how these processes directly interact with depressive symptoms. Therefore, the current literature remains largely at the level of associations and contains areas that require further development with respect to mechanistic explanations.

In conclusion, the relationship between short video-based social media use, dopamine signaling, and depressive symptoms has a multifactorial, dynamic, and context-dependent nature. Although the literature generally demonstrates significant associations, the strength and direction of this relationship vary depending on study designs, measurement instruments, and individual differences. Therefore, supporting future research with longitudinal study designs, neuroimaging methods, and more homogeneous sample groups is of critical importance for reducing the existing knowledge gaps in this field.

5. Conclusion

This narrative review examined the relationship between short video-based social media use, dopamine signaling, and depressive symptoms in light of the current literature. The findings obtained indicate that short video platforms may influence the dopaminergic reward system through their variable reward structure, algorithmic content feed, and continuous presentation of stimuli, and that this may be associated with behavioral addiction-like patterns.

The literature particularly points to the neurobiological basis of short video use through the reward prediction error mechanism and reward system activation. However, the fact that these relationships are largely explained through indirect measurements and theoretical models indicates that caution should be exercised in the interpretation of the current evidence.

When evaluated in terms of mental health, short video use generally shows negative associations with depressive symptoms, anxiety, and psychological well-being; however, this effect appears to be determined not only by the duration of use but also by the pattern of use, individual characteristics, and the social context. Therefore, the effects of short video use cannot be explained within a one-dimensional framework.

Overall, this review demonstrates that the relationship between short video-based social media use, the dopamine system, and depressive symptoms has a multifactorial and indirect structure. However, the lack of evidence regarding causality, longitudinal follow-up data, and neurobiological validation remains a major limitation in the current literature.

Future longitudinal and experimental studies investigating these relationships by combining neuroimaging techniques with long-term follow-up data will contribute to a more reliable understanding of the effects of digital media use on mental health.

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Conflict of Interest Statement

No conflict of interest has been declared.